

## CS UNIT-2 IMPORTANT ANSWERS

### 1) What are smishing and vishing?

- **Smishing (SMS Phishing):** A cyberattack where attackers send fraudulent SMS messages, pretending to be a trusted entity, to trick users into revealing personal or financial information.
- **Vishing (Voice Phishing):** A social engineering attack where attackers make phone calls pretending to be from banks, tech support, or government agencies to deceive victims into sharing sensitive data or making payments.

### 2) Why are the protocols HTTP, DNS, and ICMP suitable for attackers to use network tunneling for attacks?

- **HTTP:** Often allowed through firewalls, making it easy for attackers to hide malicious data within normal web traffic.
- **DNS:** Can be used to tunnel data by encoding information inside DNS queries and responses, bypassing security measures.
- **ICMP:** Designed for diagnostic purposes (e.g., ping requests), but attackers can use it to covertly transmit data without detection.

### 3) How do attackers use vishing to steal personal information?

- Attackers call victims, pretending to be from legitimate institutions (banks, government agencies, or tech support).
- They create urgency (e.g., "Your bank account is at risk!") to pressure victims into revealing confidential data.
- Victims may disclose sensitive details like credit card numbers, passwords, or OTPs, which attackers exploit for fraud.

### 4) Explain how attackers can use network tunneling and the HTTP protocol for conducting attacks.

- **Network Tunneling:** Attackers encapsulate malicious traffic inside legitimate protocols (like HTTP) to bypass security measures.
- **Using HTTP:** Attackers send encoded data inside normal web traffic (e.g., HTTP requests) to avoid detection. Firewalls and security tools often overlook such traffic, allowing attackers to exfiltrate data or control compromised systems remotely.

**5) List any four tricks played by attackers to cover their misdeeds.**

- **Log Deletion:** Attackers erase system and security logs to remove evidence of their activities.
- **File Timestamp Modification:** They alter file timestamps to confuse forensic investigations.
- **Using Proxy Chains & VPNs:** Attackers route traffic through multiple servers to hide their real location.
- **Disabling Security Tools:** They disable antivirus software and firewalls to avoid detection.

**6) What is network tunneling?**

- Network tunneling is a technique where data is transmitted within another protocol to bypass security restrictions. Attackers use it to sneak malicious traffic past security defenses, often using HTTP, DNS, or ICMP protocols.

**7) Explain how attackers use “Pay-per-click” advertisements to fetch money illegally.**

- Attackers create fake or malicious ads and use **click fraud** techniques, where bots or tricked users repeatedly click on these ads to generate revenue.
- They also use **malvertising**, where they inject malware into ad networks, infecting users who click on or view the ads.
- Fraudsters may **hijack ad revenue** by redirecting traffic from legitimate sites to their own, stealing income meant for real advertisers.

**8) Why are the protocols HTTP, DNS, and ICMP suitable for attackers to use network tunneling for attacks?**

Attackers exploit these protocols to tunnel malicious traffic and bypass security controls:

- **HTTP:** Web traffic is generally trusted and allowed by firewalls, enabling attackers to hide command-and-control (C&C) communications inside HTTP requests.
- **DNS:** By encoding malicious data within DNS queries and responses, attackers can exfiltrate sensitive information without detection.
- **ICMP:** Designed for network diagnostics, ICMP packets can be manipulated to secretly transfer data, evading traditional security measures.

9) Differentiate between antivirus and rogue antivirus.

| Feature          | Antivirus                                  | Rogue Antivirus                                                 |
|------------------|--------------------------------------------|-----------------------------------------------------------------|
| Purpose          | Detects and removes malware                | Pretends to be security software but spreads malware            |
| Legitimacy       | Created by reputable cybersecurity firms   | Developed by cybercriminals                                     |
| Warning Messages | Genuine alerts for real threats            | Fake alerts to scare users into paying for non-existent threats |
| Impact on Users  | Protects against malware and cyber threats | Steals money, personal data, or infects devices                 |

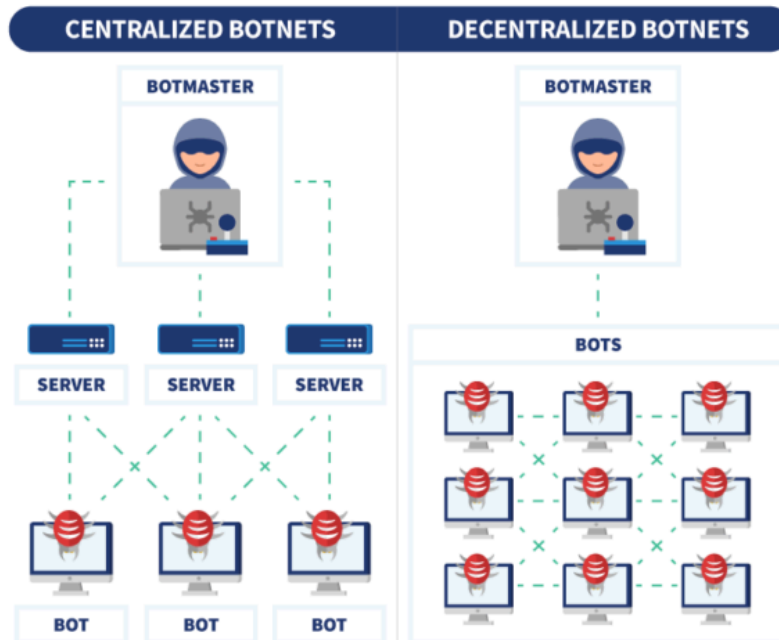
10) How does the website “whocallsme.com” help organizations understand whether customers are suspicious about their services?

- *WhoCallsMe.com* is a crowd-sourced platform where users report unknown or suspicious phone numbers.
- Organizations can monitor complaints and feedback about their customer service or marketing calls.
- If customers associate a company’s number with fraud, the organization can take corrective actions.
- It helps companies detect impersonation scams where fraudsters misuse their brand name.

11) What is a bot? Explain centralized and decentralized botnet infrastructure.

A **bot** is a compromised device infected with malware that is remotely controlled by an attacker.

- **Centralized Botnet:**
  - Bots communicate with a single **Command & Control (C&C) server**.
  - The attacker sends instructions to all infected devices via this server.
  - Easier to manage but also easier to detect and shut down.
- **Decentralized Botnet (P2P Botnet):**
  - Bots communicate with each other instead of a central server.
  - Harder to detect and take down since there’s no single point of failure.
  - Attackers use **peer-to-peer (P2P) networking** to distribute control.



## 12. How do attackers use vishing to steal personal information?

- Attackers **impersonate** banks, government agencies, or tech support over phone calls.
- They create **urgency or fear**, claiming account issues, unpaid bills, or security breaches.
- Victims are tricked into **revealing sensitive data** such as PINs, credit card numbers, or passwords.
- Attackers may use **caller ID spoofing** to appear as a legitimate institution.
- Stolen data is used for fraud, identity theft, or financial theft.

## 13. What are phishing and smishing?

- **Phishing:**
  - Cybercriminals send **fake emails** pretending to be from trusted organizations (banks, PayPal, etc.).
  - The email contains **malicious links** leading to fake login pages where victims enter credentials.
- **Smishing (SMS Phishing):**
  - Attackers send **fraudulent SMS messages** with malicious links or scam warnings.
  - Users may be tricked into revealing sensitive data or installing malware.

## 14. What is click fraud? Explain how attackers use “Pay-per-click” advertisements to fetch money illegally.

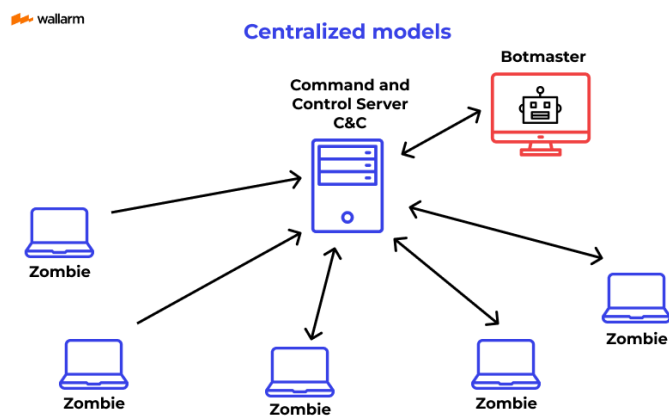
Click fraud is a technique where **fraudulent clicks** are generated on online advertisements to drain marketing budgets or earn illicit revenue.

- **How attackers exploit PPC ads:**
  - Use **bots or click farms** to repeatedly click on ads, making advertisers pay without real customer interest.
  - **Malvertising:** Insert malware into online ads to infect users' devices.
  - **Affiliate Fraud:** Trick advertisers into paying commissions on fake clicks or leads.

## 15. What is a bot? Explain centralized botnet infrastructure with a diagram.

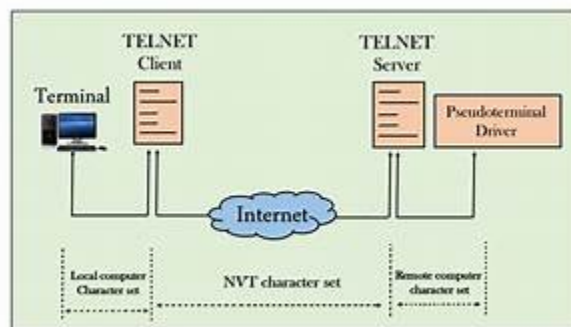
A **bot** is a malware-infected device controlled remotely by cybercriminals.

- **Centralized Botnet Infrastructure:**
  - All bots are controlled by a single **Command & Control (C&C) server**.
  - The attacker issues commands via this server to launch DDoS attacks, send spam, or steal data.
  - Easier for attackers to manage but also more vulnerable to takedowns.



**16. What is network tunneling? Explain, with a diagram, how network tunneling provides secure communication between a Telnet Client and Telnet Server. How do HTTP, DNS, and ICMP protocols help attackers bypass security measures?**

- **Network tunneling** is a technique where one protocol is encapsulated inside another to bypass security restrictions.
- **Telnet over Secure Tunnel:**
  - A **VPN or SSH tunnel** can be used to encrypt Telnet traffic, securing communication between the client and server.
- **Attackers' Use of HTTP, DNS, and ICMP for Bypassing Security:**
  - **HTTP:** Encodes malicious commands in web traffic to avoid detection.
  - **DNS:** Uses DNS queries to secretly send data outside the network.
  - **ICMP:** Misuses ping requests to transmit hidden data.



**17. Malicious code authors install local proxies. What challenges does this pose for security administrators?**

- **Traffic Obfuscation:** Attackers can use local proxies to encrypt traffic, making it harder to detect malicious activity.
- **Bypassing Security Controls:** Local proxies allow attackers to **redirect traffic** and evade firewalls.
- **Man-in-the-Middle (MITM) Attacks:** Attackers intercept and modify data passing through a local proxy.
- **Difficult Detection:** Since local proxies may look like normal services, security teams struggle to identify them.
- **Persistence Mechanism:** Even if malware is removed, the local proxy might persist, keeping the system vulnerable.

